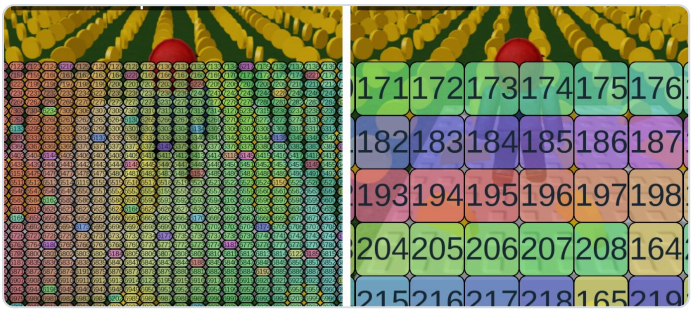
Performance and Release

Performance engineer, release manager

`scenario-roblox-performance`

Finds what makes a Roblox game slow, leaky or crash-prone, proves each fix with before and after numbers against low-end phone budgets, and prepares safe releases.



Live capture, Studio 0.743: lab before and after all fixes; server script 12.0 to 0.36 ms per frame.



Budget card for a low-end phone, the skill's default budgets. Peter McNeill, RDC 2024, at 00:36:03.

USE IT WHEN

- My game lags on mobile, stutters, or the server heartbeat drops under load
- Memory keeps growing, phones crash, or joins take too long
- Before shipping an update: restart plan, rollback path and budget check

WHAT THE EXPERTS INSIST ON

- Most crashes are out-of-memory ejections: phones eject past about 66% of 2 GB; Studio's memory number is incorrect, so test live.
  Peter McNeill (Roblox performance engineer)
- Gate on frame-time consistency: fifty-nine 10 ms frames plus one 410 ms frame still stutter at 60 fps. Andrew (Roblox Learn)
- Keep server-only data in Luau tables: tags and attributes on every part pushed one game's crash rate from 3-4% to 45%.
  McNeill and Lynn (Roblox, RDC 2025)
- A leak is a rising floor: snapshot 3 or 4 times in the same state and trust only rows that keep rising. Crusherfire (Roblox scripting educator)
- A rollback is restore, then publish, then restart outdated servers: restoring a version alone publishes nothing. Roblox Creator Docs (Version History)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Reproduce under load.** **Gate:** Baselines saved for Server and Client; the side failing the frame gate named.
2. **Audit statically.** **Gate:** Every warning fixed or justified in one line; counts inside the budget table.
3. **Name the cause.** **Gate:** Top function or profiler label named with its milliseconds per frame.
4. **Fix one thing per playtest.** **Gate:** The touched metric improves past the 5% noise floor; nothing else gets worse.
5. **Soak for leaks.** **Gate:** Memory categories and unparented instances stay flat over a 20 s soak.
6. **Check scene and streaming.** **Gate:** Draw calls, triangles and instances inside budget at the worst view.
7. **Phone check, then release.** **Gate:** Real-phone numbers or "Studio only"; restart handler tested; the owner publishes.

KEY NUMBERS

Scene budget, low-end phone (McNeill)	**500 draw calls, 500,000 triangles, 150,000 instances**
Client memory on a 2 GB phone	**under 1.3 GB; ejections start past about 66%**
Replication downstream per client	**under 50 KB/s; each server-moved assembly about 1 KB/s**
Frame gate at 60 fps	**p95 at most 20 ms, worst frame at most 33.3 ms**
Client crash rate after an update	**investigate above 2 to 3%**
Lab, all fixes (Studio 0.743)	**server script 12.0 to 0.36 ms/frame; sends 6,526 to 11 KB/s**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- Frame gate on the side that hurts: p95 and worst frame from PerfProbe, never average fps.
- compare() rows past the noise floor for each fix, one result file per variant.
- Soak flat for memory and unparented instances; every per-frame system carries a profiler label.
- Worst-view captures before and after, read on one contact sheet.

HANDOFFS

Receives Environment counts and streaming modes, Luau benchmarks, NPC bench numbers, effect counts and network payload sizes.

Delivers Leaking function to luau, release go to tooling, restart plan to data, failed budgets to UI and assets.

TOOLKIT `RBX_PERF.PY`

- `measure()`: frame, physics and network window on Server and Client
- `compare()`: per-metric before and after, past a 5% noise floor
- `script_profile()`: top functions by self time from the Script Profiler

MAIN EXPERT SOURCES

Peter McNeill (MrChickenRocket)
Roblox performance engineer, shipped Fruit Ninja, Using Roblox Tools to Improve Performance, RDC 2024

Peter McNeill and Lynn
Roblox performance engineer and Studio product manager, Top Game Performance Tales, RDC 2025

Crusherfire
Roblox scripting educator, Your Roblox Game LAGS (and how to fix it)

Andrew, Roblox Learn
Roblox official tutorial channel, How to Use the MicroProfiler

STATUS

Tests 12 Lune tests pass

In app 25 live tests pass, 4 skipped, in Roblox Studio 0.743

G11 performance

without skills	10/15
with skills	15/15